dedicated servers. Cloud-based deep learning services, like those provided by Google, Amazon, Microsoft and other players, may also be used to model a portion of the system. Fine-tuning and additional training of the system can be carried out over Edge computers.

The trained model is then used by the system in the execution phase. Based on data inputs fed to the system and referring to trainings, it can automatically provide necessary solutions. However, for time-critical applications, higher processing power is required. The same goes for large-scale video surveillance systems, where city-level infrastructure organisers are needed to strategise these investments.

The other necessity that will drive the implementation of this technology is the use of Edge analytics. Paulsson believes that while cloud-based services are quite abundant today, the data tsunami that we are looking at due to the boom of the Internet of Things (IoT) may just as well overwhelm the extent of services available. In time-critical situations, like during an ongoing crime or an accident, where a smart camera is expected to send alerts immediately, a more dedicated and in-situ network will be required. This is where Edge devices will come in. The first round of data computation will occur in the camera itself, which will help it to react in emergency situations. Full-scale analytics on the data can be taken up by the cloud platform later and eventually provide necessary insights. As a result, cloud and Edge will work in collaborative mode to deliver the highest amount of benefit.

According to a report by IDC, the security and surveillance market in India, which stands at ₹ 20 billion at present, is growing at a CAGR of 32 per cent, a trend that is expected to continue till 2023. On the other hand, the global AI and cognitive systems market is projected to cross US$ 47 billion by 2020. The mix of the two technologies are creating strong opportunities for both AI solution providers and surveillance systems producers, in addition to creating doorways for a partnership-based ecosystem. Better quality and reasonable cost can be ensured with a dense market of AI-powered surveillance platforms.

Indian players are taking the onus to leverage on this vast opportunity. C-DOT has recently revealed a satellite-based AI-powered surveillance technology that uses facial recognition and other features. It can be used in emergency situations (like natural calamities such as flood or earthquake), or in highly-crowded areas to identify lost persons or to detect environmental conditions and alert rescue teams on-the-fly. They are running trials and preparing to transfer the technology and make it commercial-ready. More players are directly partnering for Smart City projects to help with surveillance needs.

However, some challenges will have to be addressed during the implementation of these technologies, like ensuring a strong cyber-security system in place, having well-thought policies to differentiate surveillance from privacy breaches, and making all the hardware theft-proof and weatherproof.

—Paromik Chakraborty

2

GREEN TECH:

IMPACT OF SOLAR ENERGY
On Rural Development

The biggest change required is for governments to view mini-grids as complementary to traditional electric utilities, and not in conflict with them

The seventh UN Sustainable Development Goal (SDG) aims to wipe out energy poverty by 2030. Despite significant advances in electrification rates on a global level, universal access to clean electricity is far from achieved. More than one billion people around the world still live without access to electricity. India, despite being the third largest producer of electricity in the world after China and the US, was home to approximately 300 million people without access to electricity in 2017. These people mostly live in regions characterised by geographical remoteness and sparse population density, often making an extension of national electricity grid technically difficult and costly.

Established in 2015, HCL Samuday intends to develop a sustainable, scalable and replicable model—a source code for economic and social development of rural areas in partnership with central and state governments, local communities, NGOs, knowledge institutions and allied partners. It aims to improve the quality of life through optimal interventions across agriculture, education, health, infrastructure, livelihood and WASH (water, sanitation and hygiene) sectors in selected villages. HCL Samuday has committed ₹ 500 million to be spent over a five-year period in its solar program, out